PACER

Psychometric Software

LIVE WEBINAR

Getting Started with PACER

A complete psychometric environment — in your browser, on your desktop, and through Claude

Harold Doran · harold.doran@gmail.com

PACER 3.2.0 · Nexus

MY MISSION

My vision is to build the *leading, industry-grade* psychometric platform.

A comprehensive, modern environment that serves operational testing programs, researchers, and practitioners alike.

🏛️ Operational testing programs · 🔬 Researchers · 👷 Practitioners

— **Harold Doran**, creator of PACER

ONE APPLICATION BRINGS TOGETHER

- ∑ **Advanced psychometric methods**
- ⚡ **High-performance computing**
- ✨ **Emerging AI capabilities**

Making sophisticated analyses easier to **perform**, **integrate**, and **interpret**.

DISCLOSURES

An *independent* project



Independent

PACER is an independent project, not affiliated with or endorsed by any employer or organization.

TM

Trademarks

Claude, Codex, Gemma and other product names are trademarks of their respective owners. Their mention does not imply endorsement.



Views my own

The views expressed in this presentation are my own.

Harold Doran

BOOKMARK THESE

Where to *find everything*

Web app	https://www.paceronline.com
Desktop app New versions and release notes	https://github.com/hdoran/PACER_Download/releases/
Security & Compliance For IT and compliance review	https://www.paceronline.com/security.html
Guided tours	**Take a Tour** button on the landing page · **8 languages**
Template Library	https://www.paceronline.com/templates.html
PACER University	https://www.paceronline.com/university/
Technical Reference	https://www.paceronline.com/pacer-technical-reference.html
What's new	https://www.paceronline.com/changelog.html

THE NEXT 60 MINUTES

Learn the pattern once, *use it everywhere*

0–4
Why PACER
The vision, the links, the problem it solves

4–12
PACER at a glance
Modules, ways to learn, desktop, AI Advisor

12–30
Live demo: the app
IRT calibration and scoring, CDM calibration and profiling

30–48
Live demo: Claude
Optional AI that coordinates a full analysis

48–52
Get started
Your first calibration and next steps

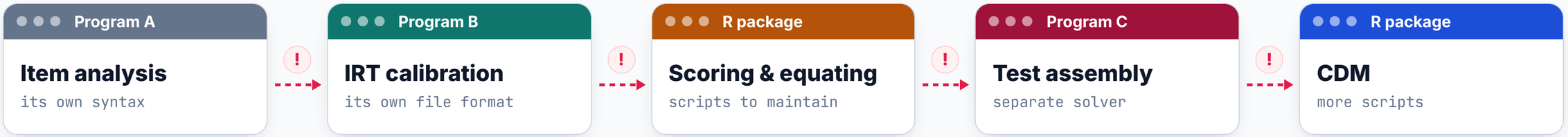
52–60
Q&A
Your questions, your data, your use cases

THE PROBLEM

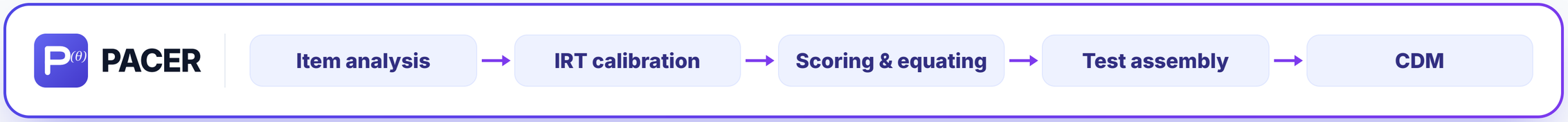
One environment, *not five*

One testing project often runs through **five different programs and R packages**, each with its own syntax, file format and license.

TODAY Every arrow is an export, a reformat and a re-import, and a new place for errors



WITH PACER Same steps, one program: results hand off automatically



✓ One program to install and learn ✓ The same five-step pattern in every module ✓ Optional AI coordinates; PACER does the math

ONE ENVIRONMENT

The full psychometric workflow, *in one place*

Item Analysis
Classical statistics, reliability, and DIF

IRT Calibration
1PL to 3PL, GRM, GPCM, RSM, multigroup, anchoring

IRT Scoring
MLE, MAP, EAP and TCC scoring with diagnostics

Equating
Stocking-Lord, Haebara, Mean/Sigma, Mean/Mean

Test Construction
Automated test assembly to a blueprint

CDM Calibration
G-DINA, A-CDM, LLM, R-RUM, DINA, DINO

Skill Profiler
Skill mastery profiles for every examinee

Latent Regression
Explanatory IRT and plausible values

- Highlighted modules are today's live demos

HELP IS BUILT IN

Six ways to *learn* PACER

Guided tours

Step-by-step walk-throughs; start from **Take a Tour** on the landing page

8 languages

FIRST CLICK

Template Library

Example files in the exact format each module expects

YOUR DATA

PACER University

Interactive lessons on IRT, CDM, equating and more

CONCEPTS

AI Advisor

Ask about the screen you're on; answers come from PACER's own documentation, locally

OPTIONAL · DESKTOP

Claude

Runs and explains whole workflows on your files

OPTIONAL · DESKTOP

Technical Reference

Every model, estimator and algorithm, with the full math

THE MATH

TWO WAYS TO RUN PACER

Same engine, *two front doors*

Web app

https://www.paceronline.com

Nothing to install. Open a browser and start analyzing.

- ✓ Every analysis module
- ✓ Guided tours, templates, University
- ✓ Ideal for trying PACER and for teaching
- – Data is uploaded to the server

BEST FOR a first look, classrooms, quick checks

Desktop app

Windows · macOS · GitHub releases

Recommended

Everything in the web app, running on your own machine. Each release ships with notes on GitHub.

- ✓ Every analysis module
- ✓ Native engine — **fast**
- ✓ Runs **100% air-gapped** — no internet needed
- ✓ Data **never leaves** your computer
- ✓ **Optional** AI Advisor and Claude Desktop

BEST FOR real projects, protected data, AI workflows

WHY A DESKTOP APP

Three reasons to *run it locally*

Speed

The estimation engine runs natively on your machine, in parallel across your processor cores. No uploads, no waiting on a server.

LSAT 2PL, 1,000 × 5 items: 0.01 s estimation

AI, if you want it

A local AI Advisor that knows PACER's documentation, plus a one-click connection that lets Claude run PACER analyses. Both are always optional — off until you turn them on.

Always optional

100% air-gapped

Every analysis runs with no internet connection. Response files are read on your computer and never uploaded — essential for protected student and candidate data.

No network, no upload

OPTIONAL · DESKTOP APP

An AI Advisor *grounded in* PACER

A retrieval-augmented generation (RAG) pipeline built from PACER's own documentation — not a general-purpose chatbot. Like any AI, check what it tells you.

1 · ASK
Your question
Asked from any page; the Advisor knows which module you're on

→

2 · RETRIEVE
PACER knowledge
~1,500 passages from the docs, technical reference and screen guides

→

3 · GENERATE
Gemma 3 (4B)
Google's open-weights model writes the answer from those passages

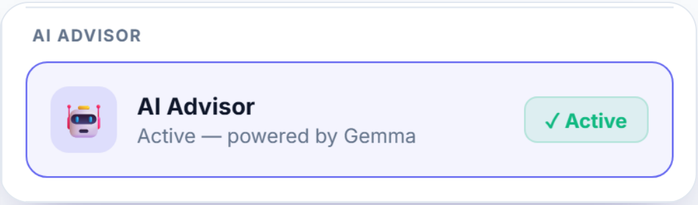
→

4 · ANSWER
PACER-specific help
Options, outputs and next steps for the screen you're on

🧩
Optional
Settings → AI Advisor → Set Up. Skip it and PACER is unchanged.

💻
Installed locally
Runs through Ollama on your computer; one-time ~3 GB download.

🔒
Stays offline
After setup it needs no internet and sends nothing anywhere.



01

PART ONE · LIVE DEMO

The PACER app

📈 IRT Calibration

🎯 IRT Scoring

🧠 CDM Calibration

🗺️ Skill Profiler

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THE ONE THING TO REMEMBER

Every module follows *the same five steps*



STEP 1
Upload
Drop a CSV or Excel file; PACER checks it instantly

STEP 2
Configure
Choose the model and options in the sidebar

STEP 3
Run
One button; results in seconds

STEP 4
Read
Summary cards, tables and interactive plots

STEP 5
Export & hand off
Download CSVs, or send results to the next module

💡 Once you have seen it twice, you know **Item Analysis, Equating and Test Construction** too — they work exactly the same way.

DEMO 1

IRT *calibration and scoring*

IRT Calibration `LSAT.csv · 1,000 × 5`

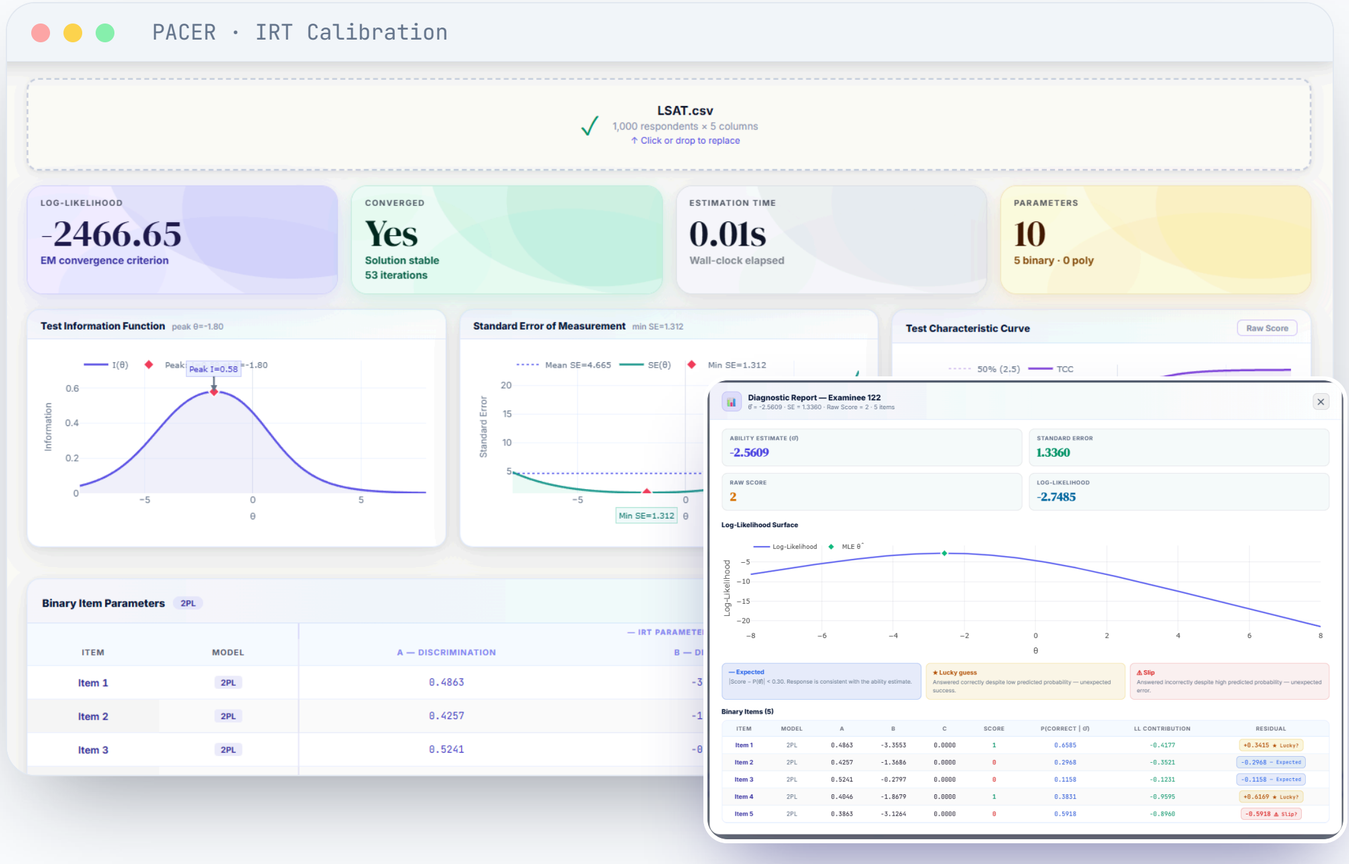
1. Upload the response matrix
2. Choose a 2PL in the sidebar
3. Run; the TIF peaks at θ = −1.8: an easy test
4. Open an item curve and export

IRT Scoring `carries over`

1. Parameters and data load automatically
2. Score with EAP
3. Open one examinee's diagnostic report

REMEMBER THESE FOR LATER

`Item 1: a = 0.4863 · b = -3.3553`



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DEMO 2

CDM calibration and *skill profiles*

gdina_responses.csv + Q-matrix · 10 items · 3 skills · 1,000 examinees

SATURATED

G-DINA

ALL / ANY SKILL

DINA DINO

NO INTERACTIONS · NEW

A-CDM LLM R-RUM

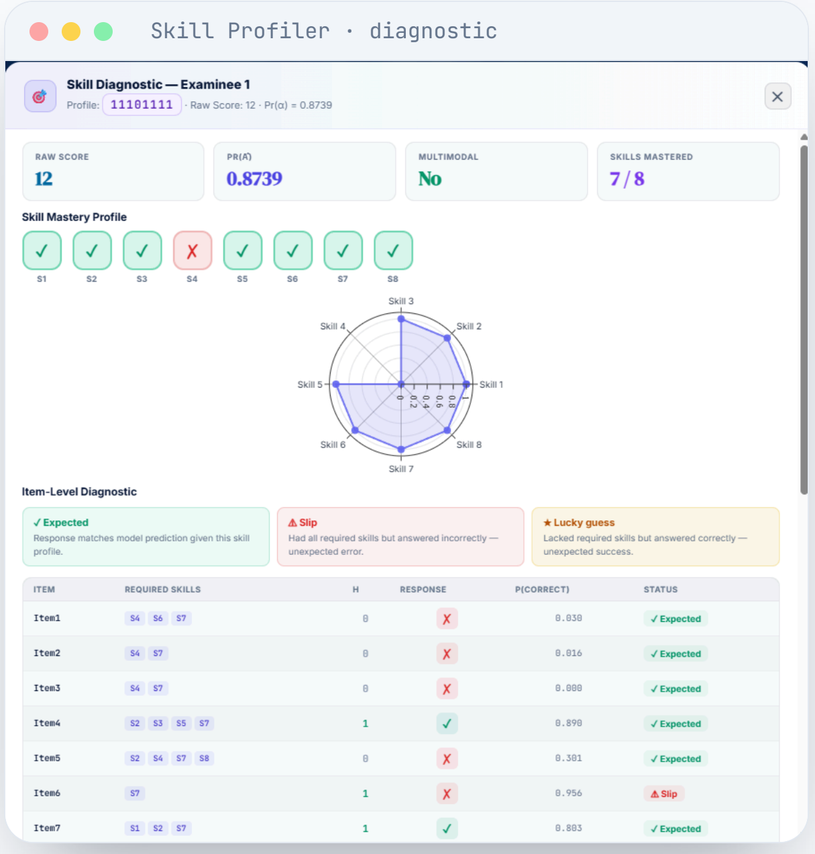
Additive on the probability, logit or log scale

Do we need the most complex model?

BIC · lower is better

R-RUM	12416
G-DINA	12420
LLM	12434
A-CDM	12481
DINA	12507
DINO	12637

A 4-point BIC gap between R-RUM and G-DINA is a near tie: the simpler model competes.



02

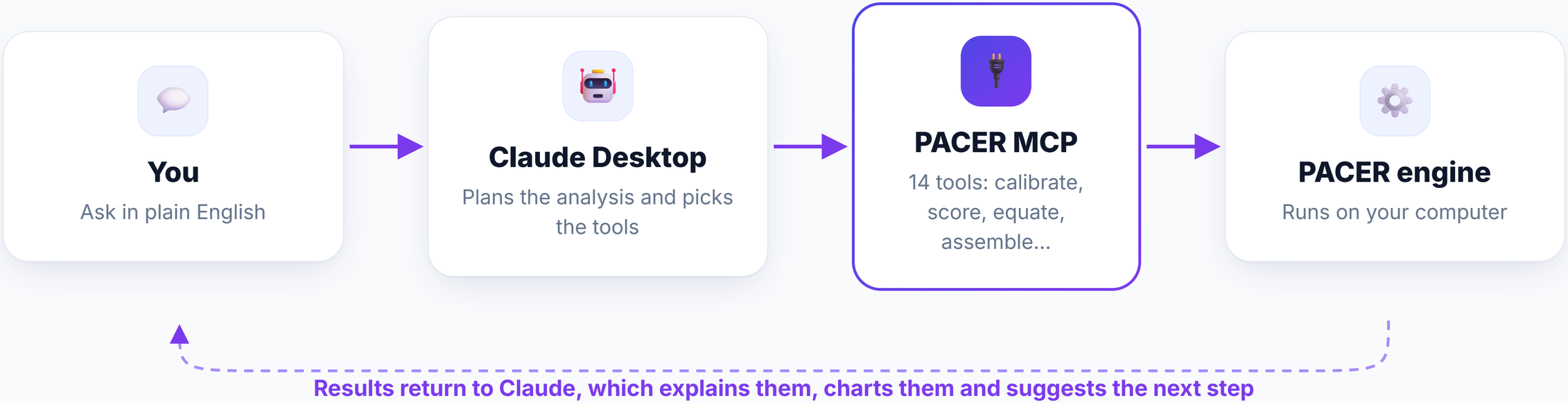
PART TWO · LIVE DEMO

PACER through Claude

The same engine and the same answers — requested in plain English and explained back to you.

HOW IT WORKS

Claude asks, PACER *computes*



🔗 You need the **PACER desktop app** plus **Claude Desktop** — or any MCP-capable desktop or command-line assistant.

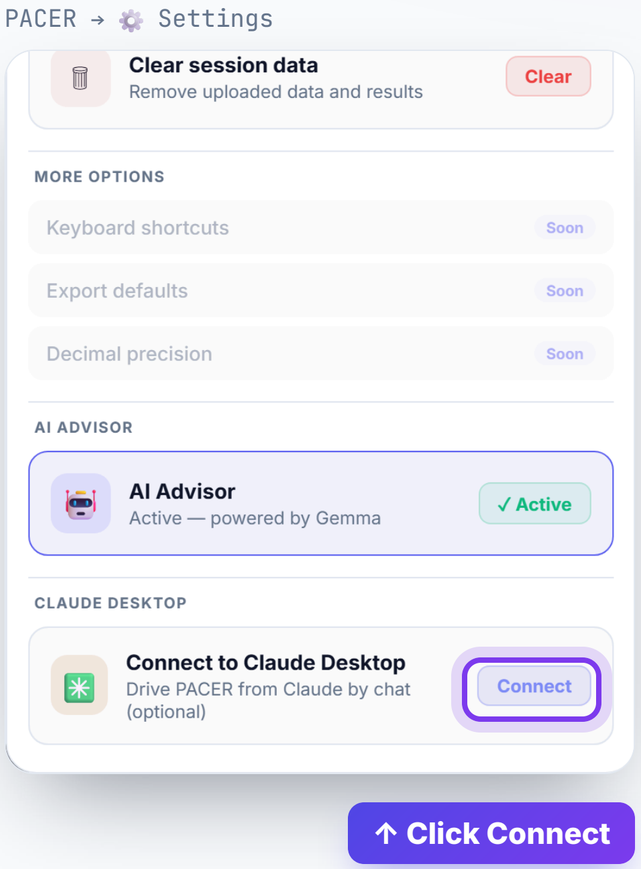
ONE-TIME SETUP

Connect Claude *in one click*

YOU NEED

The **PACER desktop app** and **Claude Desktop** — or another MCP-capable desktop or command-line assistant (setup in the appendix).

1. Click ⚙ in PACER's top bar to open **Settings**
2. Under **Claude Desktop**, click **Connect**
3. Restart **Claude Desktop** (click Connect again after each PACER upgrade)
4. Ask: `What PACER tools do you have?`



DATA PROTECTION WITH CLAUDE

Where your data *actually goes*

🖥️ YOUR COMPUTER

DATA FILES

Responses, Q-matrix, item parameters

Read from disk, never uploaded

PACER ENGINE

Every calibration and score

Computed on your processor

AI ADVISOR

Local language model

Optional; answers offline, sends nothing

CLAUDE DESKTOP

Calls PACER tools locally

Optional; receives your prompts and the results its tools return

Your prompts + tool results

Anthropic

Only if you choose Claude. Response files are not uploaded, but prompts and returned results can still contain sensitive details. Without Claude, PACER stays 100% air-gapped.

WHAT WE'LL ASK

Four prompts, *one conversation*

1 DISCOVER

```
What PACER tools do you have?
```

2 CALIBRATE

```
Calibrate a 2PL on LSAT.csv.
```

3 SCORE

```
Score the examinees with EAP and summarize the
distribution.
```

4 FULL ANALYSIS

```
Using my response file and Q-matrix, compare G-DINA
with the simpler CDMs, flag problem items, profile
the examinees, and write a short technical report.
```

TRUST, BUT VERIFY

Same engine, *same answer*

In the PACER app

ITEM 1 · a **0.4863**

ITEM 1 · b **-3.3553**

=

Through Claude

ITEM 1 · a **0.4863**

ITEM 1 · b **-3.3553**

Claude never estimates anything itself. It calls PACER's engine, so with the **same data, model and settings** the numbers match the app — and Claude then explains them.

THANK YOU

Questions?

Download PACER. Run your first calibration. See what's possible.

CONTACT

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TRY IT TODAY

https://www.paceronline.com

Desktop: github.com/hdoran/PACER_Download/releases

SCAN TO VISIT

FOR REFERENCE AFTER THE WEBINAR

Appendix

Cheat sheets, setup steps and answers to the questions people ask after their first session.

A Quick start: your first hour p. 24
B Data file cheat sheet p. 25
C Choosing an IRT model and method p. 26
D Choosing a CDM p. 27
E Prompt library for Claude p. 28
F Claude setup and troubleshooting p. 29
G Config for Codex, Cursor and others p. 30
H Security and data FAQ for IT p. 31
I How PACER is verified p. 32
J Installing, updates and support p. 33
K Glossary p. 34

APPENDIX A · QUICK START

Your *first hour* with PACER

1 Get PACER

Use the web app at `https://www.paceronline.com`, or install the desktop app from GitHub releases.

2 Take a Tour

Click **Take a Tour** on the landing page for a guided walk-through, in 8 languages.

3 Grab a sample file

Template Library → IRT Calibration → Response Matrix → **Download**.

4 Calibrate

Open **IRT Calibration**, drop in the file, choose **2PL**, click **Run**.

5 Read and export

Check the parameter table, item fit and the item curves; export the parameters as CSV.

6 Score

Send the results to **IRT Scoring**, choose **EAP**, run and export the scores.

💡 Stuck? Ask the **AI Advisor** on any page of the desktop app, or work through **PACER University** at `https://www.paceronline.com/university/`

APPENDIX B · DATA FILES

Data file *cheat sheet*

MODULE	FILE	LAYOUT / COLUMNS	KEY RULE
Item Analysis · IRT Calibration	Response matrix	Wide: one row per examinee, one column per item	Header = item names · 0/1 or 0–K · blank or NA = missing · optional **group** column (DIF)
Item Analysis (multiple choice)	Answer key	`item, key`	Pairs with raw A–D responses to enable distractor analysis
IRT Calibration	Anchor file	`item, param, value`	Fixes known a, b, c or d1… values before calibration
IRT Scoring	Parameters + responses	`columnName, model, a, b, c, d1…` `testID, key, score`	Loaded automatically after a PACER calibration
Equating	Item bank + new form	`item_id, model, a, b, c`	Shared **item_id** values become anchors · a **[POLYTOMOUS]** row splits mixed files
CDM Calibration	Responses + Q-matrix	0/1 matrix · items × skills	Q-matrix rows match response columns **by position**
Skill Profiler	CDM parameters	`itemName, guess, slip` `item, model, pattern, p_success`	Guess/slip for DINA and DINO; pattern file for the other four
Test Construction	Item bank	`itemid, model, a, b, c, d1…`	Extra columns become blueprint dimensions · **enemy_set**, **group_id**
IRT Regression	Responses + parameters	Wide, with a person ID · `item_id, model, a, b, c`	Columns that are not items become covariates

📁 Every format, with example files to download: `https://www.paceronline.com/templates.html`

APPENDIX C · IRT

Choosing a *model and method*

CALIBRATION MODELS

1PL / Rasch Binary	All items assumed equally discriminating; smallest samples
2PL Binary	Items differ in how sharply they separate examinees
3PL Binary	Multiple choice with guessing; needs larger samples
GRM Ordered categories	Rating scales and rubrics, modeled cumulatively
GPCM Ordered categories	Partial-credit scoring with item-specific slopes
RSM Ordered categories	One shared rating scale across all items (Likert)

Binary and polytomous items can be mixed in one file and one calibration.

SCORING METHODS

MLE	Maximum likelihood; no prior. Not finite for all-correct or all-wrong patterns.
MAP	Posterior mode; a prior keeps every score finite.
EAP	Posterior mean; stable for short tests and extreme patterns.
TCC	Converts a number-correct score to θ through the test characteristic curve.

EQUATING AND LINKING

Stocking-Lord and **Haebara** match whole characteristic curves and are the usual choice. **Mean/Sigma** and **Mean/Mean** use parameter moments and make quick cross-checks.

APPENDIX D · CDM

Choosing a *CDM*

G-DINA

2^k params

SATURATED

Every combination of required skills gets its own success probability. Fit it first; the others are special cases.

DINA

2 params

CONJUNCTIVE

Needs **all** required skills. Guess and slip only.

DINO

2 params

DISJUNCTIVE

Needs **any one** required skill. Guess and slip only.

A-CDM

k + 1 params

ADDITIVE · IDENTITY LINK

Each skill adds a fixed amount to P(correct).

LLM

k + 1 params

ADDITIVE · LOGIT LINK

Linear logistic model: each skill adds to the log-odds.

R-RUM

k + 1 params

ADDITIVE · LOG LINK

Each skill multiplies P(correct) by a fixed factor.

A simple workflow: fit G-DINA, then the simpler models, and compare them on AIC and BIC. PACER reports deviance, AIC, BIC, CAIC, AICc and SABIC for all six.

k is the number of skills an item requires. For the additive models, the option **Enforce monotonicity** keeps every skill's effect non-negative.

APPENDIX E · CLAUDE

A prompt *library*

CHECK A FILE	Check the format of C:\Data\responses.csv before I calibrate it.
DIF	Run a DIF analysis using the group column and list items with large DIF.
PLOT	Plot the item characteristic curves for the five most discriminating items.
EQUATE	Link newform.csv to itembank.csv with Stocking-Lord and report the A and B constants.
REGRESSION	Fit a latent regression of θ on the covariates in my file and draw five plausible values.
ITEM ANALYSIS	Run a classical item analysis on responses.csv and flag items with low point-biserials.
CALIBRATE	Calibrate a 3PL on responses.csv and show me the items with poor fit.
SCORE	Score the examinees with EAP and MLE and compare the two.
ASSEMBLE	Build a 40-item form from itembank.csv that meets blueprint.csv and maximizes information at θ = 0.
CDM	Fit G-DINA and R-RUM to my responses and Q-matrix, compare them on BIC, and profile the examinees.

💡 Give Claude the **full path** to each file. Ask it to **show the settings it used**, so you can repeat the run in the app and get the same numbers.

APPENDIX F · CLAUDE

Setup and *troubleshooting*

1. Install the PACER desktop app and Claude Desktop
2. In PACER, click ⚙ → **Settings** → Claude Desktop → **Connect**
3. Quit and reopen Claude Desktop
4. Ask: `What PACER tools do you have?`

CONNECT WRITES THIS FILE

```
Windows: %APPDATA%\Claude\claude_desktop_config.json
macOS: ~/Library/Application Support/Claude/
       claude_desktop_config.json
```

The Microsoft Store version of Claude Desktop is handled too.

Claude does not list the PACER tools

Quit Claude Desktop completely, from the system tray or with Cmd+Q, then reopen it. Closing the window is not enough.

It stopped working after a PACER upgrade

Click **Connect** in PACER Settings again. The saved path includes the version number.

Claude cannot find my file

Give the full path, for example C:\Data\responses.csv.

Is my data uploaded?

Raw files stay on your computer. Your prompts and the results PACER returns go to Anthropic.

Using Codex, Cursor or another assistant? See Appendix G.

APPENDIX G · CODEX, CURSOR AND OTHER ASSISTANTS

One config, *any MCP assistant*

mcp config · Windows · PACER 3.2.0

```
{
  "mcpServers": {
    "pacer": {
      "command": "C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\PACER.exe",
      "args": ["C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\resources\\pacer-mcp\\pacer-mcp.mjs"],
      "env": {
        "ELECTRON_RUN_AS_NODE": "1",
        "PACER_EXE": "C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\resources\\win-x64\\IrtEstimation.exe"
      }
    }
  }
```

OpenAI Codex
Settings → Plugins → MCPs → Add; enter the same fields

Cursor · Windsurf
`~/.cursor/mcp.json`
`…/windsurf/mcp_config.json`

VS Code · Copilot
`.vscode/mcp.json`
(key "servers")

Claude Code
`claude mcp add pacer …`

Gemini CLI
`~/.gemini/settings.json`

Replace **YOU** with your Windows user name; the **app-3.2.0** folder changes with each release. On macOS the command is `/Applications/PACER.app/Contents/MacOS/PACER`. `ELECTRON_RUN_AS_NODE` makes PACER run its tool server instead of opening a window.

APPENDIX H · FOR IT AND COMPLIANCE

Security and data *FAQ*

Is our data stored?
Web app: processed in-session and never written to persistent storage. Desktop app: data never leaves the computer.

Where is the web app hosted?
Microsoft Azure App Service in the United States.

How is traffic encrypted?
TLS 1.2 or higher, with ECDHE forward secrecy and AES-GCM ciphers. Azure encrypts infrastructure storage at rest.

Can it run offline?
Yes. The Windows and macOS desktop app makes no network calls and can run fully air-gapped.

What about the AI features?
The AI Advisor runs locally. Claude is opt-in and off by default; when on, prompts and derived results go to Anthropic.

How are vulnerabilities handled?
Dependabot and NVD monitoring, static analysis on every build, OWASP ZAP before each release. Critical fixes within 30 days, high within 60.

How long is a version supported?
Each major version for 12 months, with at least 90 days' notice before end of life.

Is PACER certified?
Not currently under SOC 2 or FedRAMP. Controls align with common practice under a documented shared-responsibility model.

Full details for procurement and due diligence: https://www.paceronline.com/security.html

APPENDIX I · TRUST, BUT VERIFY

How PACER *is verified*

IRT results are benchmarked against **mirt**, a peer-reviewed R package. CDM results are checked against the **GDINA** R package.

1PL · Rasch	2PL	2PL + GPCM	Multigroup 2PL	3PL · N = 50,000
0.003	0.004	< 0.001	0.011	0.003
RMSE, difficulty	RMSE, discrimination	Max difference, all parameters	Max difference, discrimination	RMSE, discrimination
vs mirt	vs mirt	vs mirt	vs mirt	vs mirt

CDM · GDINA 2.13.1

A-CDM, LLM and R-RUM deviance agrees to within **0.01**; pattern probabilities agree to **three decimals**.

FAIR COMPARISON

Benchmarks use D = 1.0 to match mirt; PACER's default is D = 1.702. Both packages ran with default quadrature and convergence.

Every table, with the R code to reproduce it: https://www.paceronline.com/trust.html

APPENDIX J · GETTING SET UP

Installing, updates *and support*

Web app

Any modern browser, nothing to install.

`https://www.paceronline.com`

Windows

PACERSetup.exe from the GitHub releases page. Installs to your user AppData folder.

macOS

PACER.dmg for Apple Silicon and Intel. Drag PACER to Applications.

AI Advisor (optional)

Settings → AI Advisor → Set Up. Installs through Ollama; one-time ~3 GB download, offline afterward.

Claude (optional)

Needs Claude Desktop or another MCP assistant. See Appendices F and G.

Updates

New versions and release notes on GitHub releases and the changelog. Reconnect Claude after upgrading.

`https://www.paceronline.com/changelog.html`

DESKTOP DOWNLOADS
`https://github.com/hdoran/PACER_Download/releases/`

QUESTIONS AND FEEDBACK
Harold Doran · `harold.doran@gmail.com`

APPENDIX K · GLOSSARY

Terms and *acronyms*

IRT	Item response theory: models a response from the person's ability and the item's parameters.
a, b, c	Discrimination, difficulty and lower asymptote (guessing).
GRM · GPCM · RSM	Graded response, generalized partial credit and rating scale models.
TCC · TIF	Test characteristic curve; test information function.
Equating	Placing parameters from different forms on one common scale.
Plausible values	Random draws from each examinee's posterior, used for group statistics.
CDM	Cognitive diagnosis model: classifies examinees as masters or non-masters of skills.
LLM	In PACER's CDM module, the linear logistic model, not a language model.
MCP	Model Context Protocol: the open standard that lets AI assistants call PACER's tools.
θ (theta)	The latent ability or trait being measured.
1PL · 2PL · 3PL	Logistic IRT models with one, two or three item parameters.
MLE · MAP · EAP	Maximum likelihood, posterior mode and posterior mean ability estimates.
DIF	Differential item functioning: an item behaves differently for groups of equal ability.
ATA	Automated test assembly: building forms to a blueprint with an optimizer.
LLTM	Linear logistic test model: explains item difficulty from item features.
Q-matrix	Table showing which skills each item requires.
AIC · BIC	Information criteria for comparing model fit; lower is better.
RAG	Retrieval-augmented generation: the AI Advisor answers from PACER's own documentation.